Based on week 10 lecture

this is a sample exam.

Test yourself for a duration of 30 min

① $A = \begin{bmatrix} -2 & 1 & 0 & & \\ 1 & -2 & 1 & 0 & \\ 0 & 1 & -2 & 1 & \\ & \ddots & \ddots & \ddots & \ddots \end{bmatrix}$

$A = M - N$ with $M = \mathrm{diag}(A)$

Find the iteration matrix G s.t.

$$u^{k+1} = G u^{k} + c$$

(Since A is given as number, G would be found as numbers.

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② $A = \begin{bmatrix} \alpha & \beta & & 0 \\ \beta & \alpha & \beta & \\ & \beta & \alpha & \beta \\ 0 & & & \ddots \end{bmatrix}$ is tridiagonal and symmetric. $N \times N$ matrix.

eigenvalues of A are

$$\lambda_j = \alpha + 2\beta \cos \frac{\pi j}{N+1}, \quad j = 1, 2 \dots N$$

Show that the spectral radius of G is approximately $1 - \frac{\pi^2}{2N^2}$, where G is given in Q1.

(Use Taylor expansion; $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots\dots$)

(3) Consider heat equation $\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}$

discretized on $\{x_j\}_{j=1}^{N}$ as

$$\frac{du}{dt} = \frac{\nu}{\Delta x^2} \begin{bmatrix} -2 & 1 & & & \\ 1 & -2 & 1 & & \\ 0 & 1 & -2 & 1 & \\ & & \ddots & \ddots & \ddots \end{bmatrix}$$

Using Euler Implicit method Convert the above system into

$$Au = f$$

where $u = u(x_j, t_{n+1})$, $j = 1 \cdots N$

Consider a splitting as $A = M - N$

with $M = \text{diag}(A)$.

Estimate the spectral radius of G

where $u^{k+1} = G u^k + c$

and determine the number of iterations for $N = 128, 512, 2048$ and tolarence is 10^{-5}.